

Robust Consensus in Finite Multi-Agent Populations with Disruptive Individuals

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Abstract

This work studies the robustness of consensus in finite populations of agents in the presence of disruptive individuals. We compare the voter and cross-inhibition decision mechanisms, each extended with either zealots or contrarians. The four models are formalized as continuous-time Markov chains (CTMCs), and their deterministic limits are analysed through their equilibria and local stability. Exact model checking is used for small populations, while a validated Gillespie simulator enables statistical model checking for populations up to $N = 4000$. The results show that cross-inhibition is substantially more robust than the voter model, and contrarians generally disrupt consensus more strongly than zealots. In particular, zealots in the cross-inhibition model may initially increase the probability of stable consensus on the higher-quality option. Increasing the population size makes the transition between stable and unstable regimes sharper. We also observe that disruptive individuals often prevent the maintenance of consensus rather than its formation altogether.

1 Models and consensus properties

We consider a finite population of N agents choosing between two options, A and B , with qualities

$$q_A = 1.05, \quad q_B = 0.95.$$

Two collective decision mechanisms are considered. In the voter model, agents directly copy the opinion of interacting individuals. In the cross-inhibition model, agents first enter an undecided state before changing opinion.

For each mechanism, we introduce two types of disruptive individuals. Zealots maintain a fixed opinion throughout the execution, whereas contrarians dynamically evolve so as to oppose the agents with which they interact (Appendix A). This gives four models:

voter + zealots, voter + contrarians,
cross-inhibition + zealots, cross-inhibition + contrarians.

Let supp_A and supp_B denote the total support for the two options, including the corresponding disruptive agents. We define an A -majority by

$$\text{maj}_A \equiv (\text{supp}_A \geq m) \wedge (\text{supp}_A \geq \text{supp}_B + d),$$

with an analogous definition for maj_B . We use $m = 50\%$ of the population and a separation threshold $d = 10\%$. The main stability property is

$$\varphi_A(t, h) = F_{\leq t} G_{\leq h} \text{maj}_A,$$

which requires the system to reach an A -majority by time t and then maintain it continuously for at least h time units. We use

$$t = 35, \quad h = 40,$$

and define $\varphi_B(t, h)$ analogously.

2 Verification and simulation

Each model is implemented as a CTMC. Exact bounded reachability probabilities are computed with Stormpy for small and medium populations, up to $N = 200$.

To analyze larger populations, we implemented a direct Gillespie simulator that computes the enabled reactions and their rates directly from the current population state, without constructing the complete state space. Stability and reachability properties are evaluated online, so complete trajectories do not need to be stored.

The direct simulator was validated in three ways:

1. direct reachability estimates were compared with exact model-checking probabilities;
2. direct and Storm-based simulations produced statistically compatible probability estimates;
3. online monitors agreed with offline trajectory monitors on every tested trajectory.

The statistical experiments use 4500 simulations per configuration and 99% Wilson confidence intervals. We considered

$$N \in \{20, 50, 100, 200, 500, 1000, 2000, 4000\},$$

and disruptor fractions between 0 and 0.6.

To study how the persistence of consensus depends on the required holding time, we performed a separate experiment evaluating

$$\varphi_A(35, h) = F_{\leq 35} G_{\leq h} \text{maj}_A$$

for

$$h \in \{0, 5, 10, 20, 30, 40, 60, 80\}.$$

For $h = 0$, this property reduces to reaching either majority within time 35. Each trajectory was simulated only once up to time $35 + 80 = 115$, while all holding-time properties were monitored simultaneously.

For each disruptor fraction ρ , we also estimate the 50% persistence horizon

$$h_{0.5}^A = \sup \{h \geq 0 : P(\varphi_A(35, h)) \geq 0.5\}.$$

The crossing point is obtained by linear interpolation between consecutive tested values of h . When the probability remains at least 0.5 at $h = 80$, the estimate is right-censored and only $h_{0.5}^A > 80$ can be concluded.

Deterministic limit. In addition to the stochastic analysis, the density-dependent population models admit deterministic mean-field limits as $N \rightarrow \infty$. The corresponding ODE systems and their equilibrium analysis are reported in Appendix B. Analytical stability and bifurcation results are derived for the symmetric benchmark $q_A = q_B$. Since the experiments use $q_A = 1.05$ and $q_B = 0.95$, these results provide qualitative insight into the large-population structure, while the quantitative conclusions are obtained from statistical model checking (SMC).

3 Results

Figure 1a shows the probabilities of stable consensus for both options, for $N = 100$. Both voter variants rapidly lose stability as the disruptor fraction increases. Contrarians are especially effective: in the voter model, stable consensus is already almost absent at a disruptor fraction of 0.1.

Cross-inhibition is substantially more robust. With zealots, the probability of stable A -consensus initially increases, reaching its largest observed value near a disruptor fraction of 0.3, before collapsing at larger fractions. Thus, a moderate number of fixed supporters can favor the higher-quality option without immediately destroying consensus. Cross-inhibition with contrarians is less robust than its zealot counterpart, but it still tolerates substantially more disruption than the voter model.

Increasing N makes the transition between stable and unstable regimes progressively sharper. The sampled grid shown in Figure 1b identifies the following large- N transition regions:

Model	Transition region
Voter + zealots	0.25–0.32
Voter + contrarians	0.12–0.15
Cross-inhibition + zealots	0.45–0.50
Cross-inhibition + contrarians	0.24–0.28

Table 1: Approximate disruptor-fraction intervals in which the large- N probability of stable A -consensus drops sharply.

Figure 1c moreover shows that, as N increases, the transition generally moves towards larger disruptor fractions and becomes progressively sharper. For cross-inhibition with zealots, the largest populations remain close to probability one up to around 0.40–0.45, then collapse near 0.48. For cross-inhibition with contrarians, the transition becomes very sharp near 0.25–0.28.

For small N instead, the probability can be substantially below one even in absence of disruptors. This is a finite-size effect: stochastic fluctuations can either favor the lower-quality option or prevent an A -majority from being maintained for the entire required holding interval.

A comparison between simple reachability and stable consensus, reported in Figures 2a and 2b, highlights a clear separation between reaching a majority and maintaining it. For $N=100$, the population reaches some majority with probability close to one in all tested configurations, even when the probability of sustaining it is nearly zero. For $N=1000$, stable consensus still disappears well before simple reachability, although sufficiently large disruptor fractions eventually reduce the probability of reaching a majority as well. Overall, disruptive individuals primarily undermine the persistence of majority states, producing transient majorities before they begin to inhibit majority formation altogether.

Holding-time analysis. As expected, Figure 4 shows that the probability of stable A -consensus decreases monotonically as h increases, but the rate of decay strongly depends on the model and the disruptor fraction. The voter models lose persistence rapidly, with contrarians having the stronger effect: for $N = 100$ and a disruptor fraction

of $\rho = 0.1$, the probability falls below 0.5 at approximately $h = 30$ with zealots and $h = 13$ with contrarians; only the configurations without disruptors maintain probability above 0.5 throughout the complete tested interval.

Cross-inhibition instead exhibits a clearer separation between a persistent and a non-persistent regime. For $N = 1000$, $h_{0.5}^A > 80$ on the sampled grid up to $\rho = 0.36$ with zealots and up to $\rho = 0.20$ with contrarians. For $N = 100$, the corresponding values are $\rho = 0.30$ and $\rho = 0.16$. Beyond these regions, the persistence horizon drops rapidly, in some cases to only a few time units.

Increasing the population size therefore makes the transition sharper. Below the transition region, the larger systems maintain A -consensus with probability close to one for the entire tested interval; above it, stability rapidly collapses. Curves that start below one at $h = 0$ but remain approximately constant indicate that the main failure is reaching A -majority within time 35, rather than maintaining it once it has been reached.

Simulation runtime. Figure 3 shows that the simulation time per episode grows approximately linearly with the population size N , as indicated by the nearly straight curves on the logarithmic axes. The zealot variants are consistently faster, while the two contrarian variants require approximately twice as much time per episode because they include additional interaction mechanisms and state updates. Cross-inhibition with zealots is the fastest of the four models. Although the variability across disruptor fractions increases with N , it does not substantially alter the overall scaling trend. Even for $N = 4000$, the median runtime remains below one second per episode, confirming that the direct Python simulator makes statistical analysis of populations far beyond the range of exact model checking feasible.

Relation with the deterministic limit. The sharp transitions observed by SMC are consistent with the equilibrium analysis of the limiting ODE systems, particularly in the cross-inhibition models. In the symmetric case, cross-inhibition with zealots has two stable opinion-dominated equilibria below the critical total zealot fraction $2z = 0.4$; similarly, cross-inhibition with contrarians has two stable asymmetric equilibria for $C < 0.25$. Above these values, only the symmetric equilibrium remains stable, corresponding to the loss of a persistent majority.

These critical values are close to the large- N SMC transitions, observed around 0.45–0.50 for zealots and 0.24–0.28 for contrarians. The agreement is not expected to be exact, since the simulations use $q_A = 1.05$ and $q_B = 0.95$, whereas the analytical thresholds were derived for equal qualities. Moreover, the SMC property also imposes finite reaching and holding times.

For the voter models, the deterministic systems do not show the same type of bifurcation. Instead, they have a stable internal equilibrium whose position changes with the disruptor fraction. The observed SMC transition occurs when the dynamics no longer remain sufficiently far from the symmetric state to satisfy the majority condition for the required duration. As N increases, stochastic fluctuations become smaller and trajectories follow the deterministic behavior more closely, explaining why the transition becomes sharper.

4 Conclusion

The experiments confirm that the choice of interaction mechanism has a major effect on robustness. Cross-inhibition maintains stable decisions under substantially larger disruptive fractions than the voter model. Contrarians are generally more damaging than zealots, while a moderate number of zealots may improve the selection of the better option under cross-inhibition. Larger populations exhibit sharper transitions, which is qualitatively consistent with the deterministic analysis, particularly for the cross-inhibition models, whose limiting systems undergo a change in equilibrium stability near the transition fractions observed.

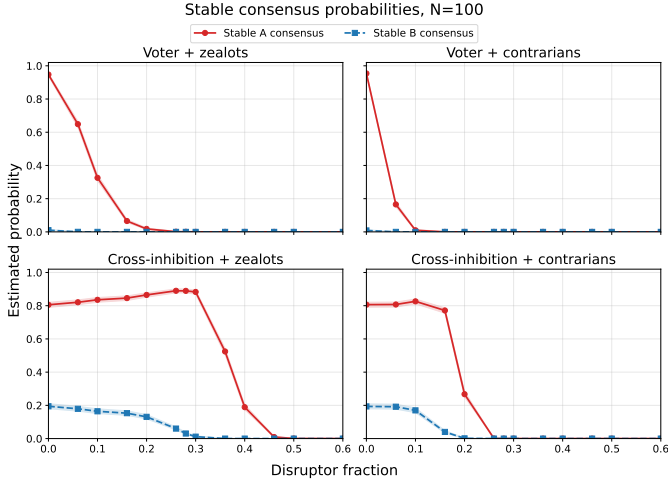
The distinction between reaching and maintaining consensus is crucial: majority states remain easy to reach even when stable consensus becomes nearly impossible. The holding-time analysis confirms that disruptive individuals affect the duration of consensus.

Overall, the results highlight the importance of distinguishing majority formation from persistent consensus and show that cross-inhibition provides the strongest robustness against both types of disruptive individuals.

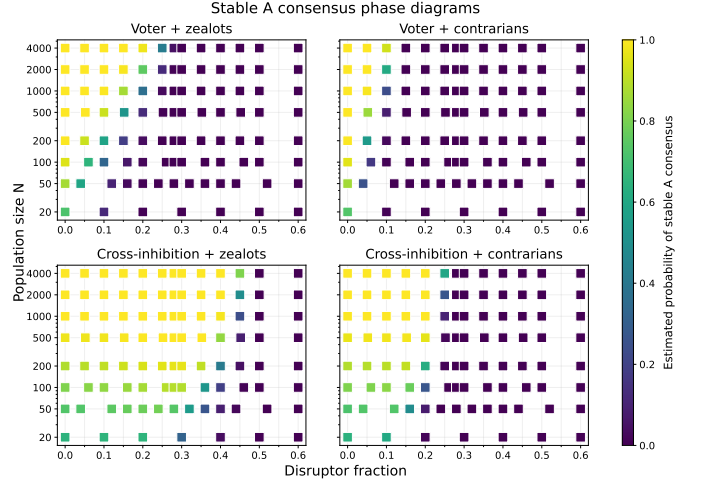
References

- [1] Luca Bortolussi, Jane Hillston, Diego Latella, and Mieke Massink. Continuous approximation of collective system behaviour: A tutorial. *Performance Evaluation*, 70(5):317–349, 2013.
- [2] Julia Klein and Tatjana Petrov. Quantifying consensus in stochastic swarms with disruptive individuals. In *2025 European Control Conference (ECC)*, pages 272–277, 2025.

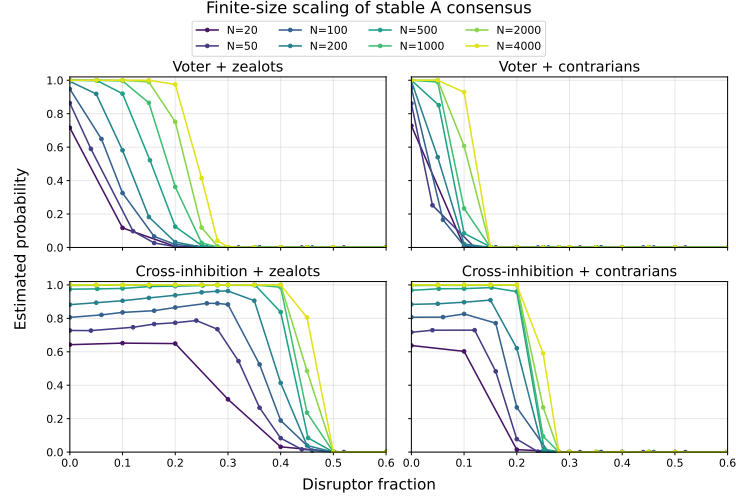
5 Figures



(a) Estimated probabilities of stable A - and B -consensus for $N = 100$.

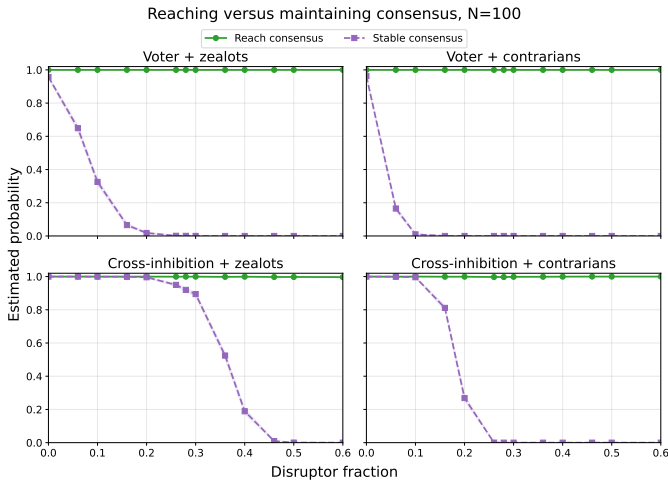


(b) Stable A -consensus sampled phase diagrams.

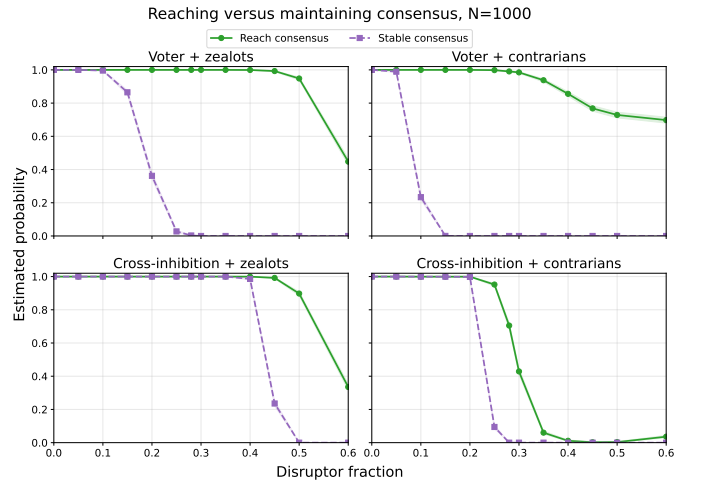


(c) Finite-size scaling of stable A -consensus probability.

Figure 1: Stable-consensus results for $q_A = 1.05$, $q_B = 0.95$, $t = 35$, and $h = 40$. Shaded regions represent 99% Wilson confidence intervals obtained from 4500 simulations per configuration.



(a) $N = 100$.



(b) $N = 1000$.

Figure 2: Comparison between reaching and maintaining consensus on either option. Shaded regions represent 99% Wilson confidence intervals obtained from 4500 simulations per configuration.

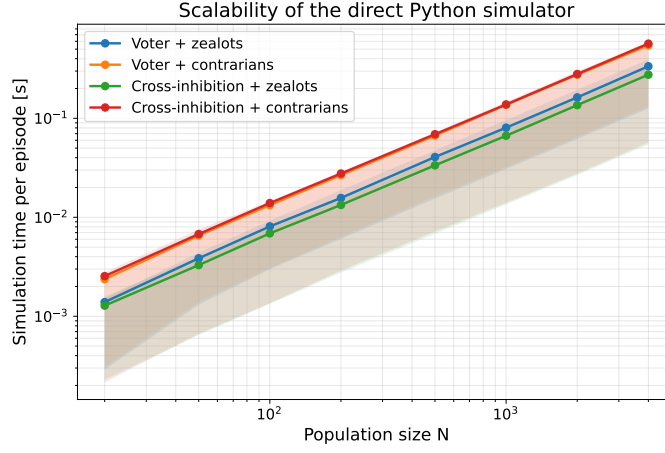
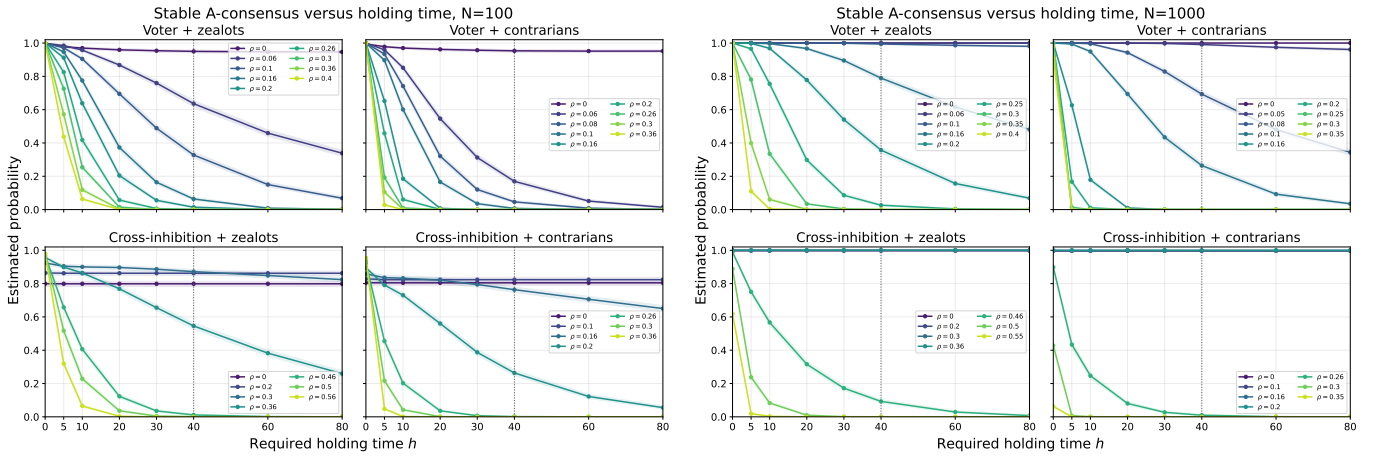
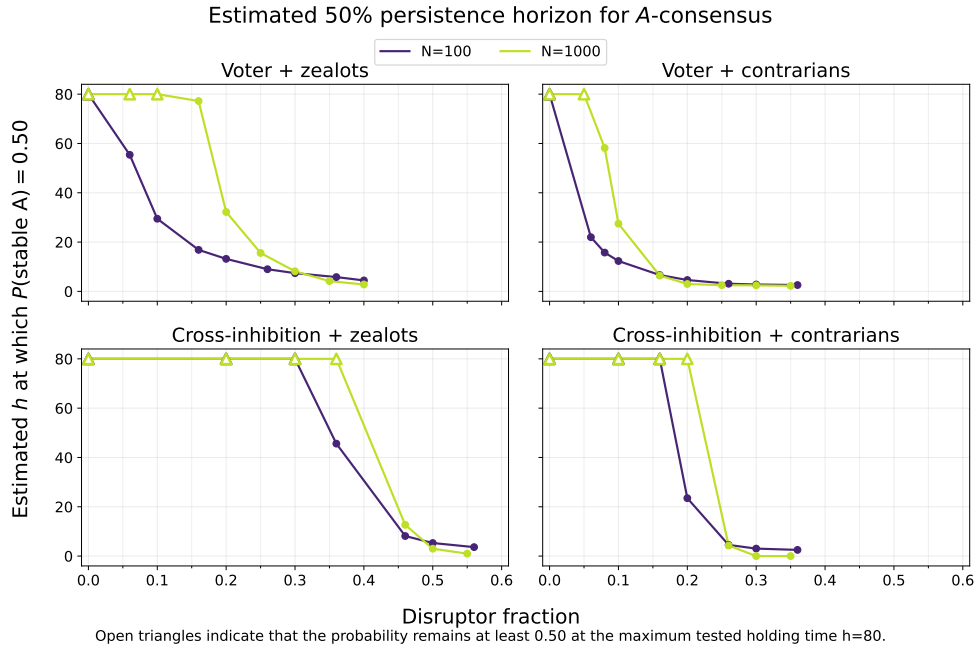


Figure 3: Median simulation time per episode as a function of N . Shaded bands show the interquartile range across disruptor fractions.



(a) Stable A -consensus versus h , for $N = 100$.

(b) Stable A -consensus versus h , for $N = 1000$.



(c) Estimated 50%-persistence horizon $h_{0.5}$.

Figure 4: Sensitivity of stable A -consensus to the required holding time. Panels (a) and (b) show the estimated probability of $F_{\leq 35}G_{\leq h} \text{maj}_A$ for different disruptor fractions ρ . Shaded regions represent 99% Wilson confidence intervals obtained from 4500 simulations per configuration. Panel (c) reports the estimated holding time $h_{0.5}$ at which this probability falls to 0.5; open triangles indicate that $h_{0.5} > 80$.

A Models

A.1 Voter model with zealots

Label	Reaction	Update vector	Rate
pure_A	$A + B \rightarrow 2A$	$(1, -1, 0, 0)$	$q_A \frac{X_A X_B}{N}$
pure_B	$A + B \rightarrow 2B$	$(-1, 1, 0, 0)$	$q_B \frac{X_A X_B}{N}$
zea_A	$B + Z_A \rightarrow A + Z_A$	$(1, -1, 0, 0)$	$q_A \frac{X_B X_{Z_A}}{N}$
zea_B	$A + Z_B \rightarrow B + Z_B$	$(-1, 1, 0, 0)$	$q_B \frac{X_A X_{Z_B}}{N}$

A.2 Voter model with contrarians

Label	Reaction	Update vector	Rate
pure_A	$A + B \rightarrow 2A$	$(1, -1, 0, 0)$	$q_A \frac{X_A X_B}{N}$
pure_B	$A + B \rightarrow 2B$	$(-1, 1, 0, 0)$	$q_B \frac{X_A X_B}{N}$
con_B_ord	$A + C_B \rightarrow B + C_B$	$(-1, 1, 0, 0)$	$q_B \frac{X_A X_{C_B}}{N}$
con_A_sw	$A + C_A \rightarrow A + C_B$	$(0, 0, -1, 1)$	$q_A \frac{X_A X_{C_A}}{N}$
con_A_ord	$B + C_A \rightarrow A + C_A$	$(1, -1, 0, 0)$	$q_A \frac{X_B X_{C_A}}{N}$
con_B_sw	$B + C_B \rightarrow B + C_A$	$(0, 0, 1, -1)$	$q_B \frac{X_B X_{C_B}}{N}$
con_A_con	$C_A + C_A \rightarrow 2C_B$	$(0, 0, -2, 2)$	$q_A \frac{X_{C_A}(X_{C_A}-1)}{2N}$
con_B_con	$C_B + C_B \rightarrow 2C_A$	$(0, 0, 2, -2)$	$q_B \frac{X_{C_B}(X_{C_B}-1)}{2N}$

A.3 Cross-inhibition model with zealots

Label	Reaction	Update vector	Rate
cross_A	$A + B \rightarrow A + U$	$(0, -1, 1, 0, 0)$	$q_A \frac{X_A X_B}{N}$
cross_B	$A + B \rightarrow U + B$	$(-1, 0, 1, 0, 0)$	$q_B \frac{X_A X_B}{N}$
rec_A	$A + U \rightarrow 2A$	$(1, 0, -1, 0, 0)$	$q_A \frac{X_A X_U}{N}$
rec_B	$B + U \rightarrow 2B$	$(0, 1, -1, 0, 0)$	$q_B \frac{X_B X_U}{N}$
zea_cross_B	$A + Z_B \rightarrow U + Z_B$	$(-1, 0, 1, 0, 0)$	$q_B \frac{X_A X_{Z_B}}{N}$
zea_rec_B	$U + Z_B \rightarrow B + Z_B$	$(0, 1, -1, 0, 0)$	$q_B \frac{X_U X_{Z_B}}{N}$
zea_cross_A	$B + Z_A \rightarrow U + Z_A$	$(0, -1, 1, 0, 0)$	$q_A \frac{X_B X_{Z_A}}{N}$
zea_rec_A	$U + Z_A \rightarrow A + Z_A$	$(1, 0, -1, 0, 0)$	$q_A \frac{X_U X_{Z_A}}{N}$

A.4 Cross-inhibition model with contrarians

Label	Reaction	Update vector	Rate
cross_A	$A + B \rightarrow A + U$	$(0, -1, 1, 0, 0)$	$q_A \frac{X_A X_B}{N}$
cross_B	$A + B \rightarrow U + B$	$(-1, 0, 1, 0, 0)$	$q_B \frac{X_A X_B}{N}$
rec_A	$A + U \rightarrow 2A$	$(1, 0, -1, 0, 0)$	$q_A \frac{X_A X_U}{N}$
rec_B	$B + U \rightarrow 2B$	$(0, 1, -1, 0, 0)$	$q_B \frac{X_B X_U}{N}$
con_cross_B	$A + C_B \rightarrow U + C_B$	$(-1, 0, 1, 0, 0)$	$q_B \frac{X_A X_{C_B}}{N}$
con_rec_B	$U + C_B \rightarrow B + C_B$	$(0, 1, -1, 0, 0)$	$q_B \frac{X_U X_{C_B}}{N}$
con_A_sw	$A + C_A \rightarrow A + C_B$	$(0, 0, 0, -1, 1)$	$q_A \frac{X_A X_{C_A}}{N}$
con_cross_A	$B + C_A \rightarrow U + C_A$	$(0, -1, 1, 0, 0)$	$q_A \frac{X_B X_{C_A}}{N}$
con_rec_A	$U + C_A \rightarrow A + C_A$	$(1, 0, -1, 0, 0)$	$q_A \frac{X_U X_{C_A}}{N}$
con_B_sw	$B + C_B \rightarrow B + C_A$	$(0, 0, 0, 1, -1)$	$q_B \frac{X_B X_{C_B}}{N}$
con_A_con	$C_A + C_A \rightarrow C_B + C_B$	$(0, 0, 0, -2, 2)$	$q_A \frac{X_{C_A}(X_{C_A}-1)}{2N}$
con_B_con	$C_B + C_B \rightarrow C_A + C_A$	$(0, 0, 0, 2, -2)$	$q_B \frac{X_{C_B}(X_{C_B}-1)}{2N}$

B Deterministic Mean-Field Analysis

Let $X_i^{(N)}(t)$ denote the number of agents in state i at time t in a population of size N , and define the corresponding normalised process by

$$\hat{X}_i^{(N)}(t) = \frac{X_i^{(N)}(t)}{N}.$$

Under the standard density-dependent scaling of the transition rates, the normalised stochastic process converges, as $N \rightarrow \infty$, to a deterministic trajectory satisfying a system of ordinary differential equations; see, for example, [1]. To simplify notation, $x_A, x_B, u, z_A, z_B, c_A, c_B$ will denote the limiting concentration associated with ordinary A -agents, ordinary B -agents, undecided agents, A -zealots, B -zealots, A -contrarians, and B -contrarians, respectively.

The analytical results are derived for arbitrary $q_A, q_B > 0$ whenever possible. More explicit equilibrium and stability results are obtained for the symmetric benchmark $q_A = q_B = q$. The numerical experiments instead use $q_A > q_B$, therefore, the symmetric analysis should be interpreted as a qualitative description of the large-population dynamics rather than as a direct prediction of the experimental probabilities.

Conservation laws For all the models considered, the total population size is conserved: $\sum_i \hat{X}_i^{(N)}(t) = 1$.

In the variants with zealots, the numbers of zealots committed to each opinion remain constant:

$$X_{Z_A}^{(N)}(t) = X_{Z_A}^{(N)}(0), \quad X_{Z_B}^{(N)}(t) = X_{Z_B}^{(N)}(0).$$

As a result, the total number of non-zealot agents is also constant and is denoted by $S^{(N)} := 1 - \hat{X}_{Z_A}^{(N)}(0) - \hat{X}_{Z_B}^{(N)}(0)$.

Similarly, in models with contrarians, the total number of contrarian agents is conserved:

$$C^{(N)} := \hat{X}_{C_A}^{(N)}(t) + \hat{X}_{C_B}^{(N)}(t).$$

It follows that the total number of ordinary agents remains constant as well, given by $S^{(N)} := 1 - C^{(N)}$.

In the limit we'll use $S = 1 - z_A - z_B$ in zealots models, $S = 1 - C = 1 - c_A - c_B$ in contrarians models.

B.1 Voter model with zealots

The deterministic limit is

$$\begin{cases} \dot{x}_A = (q_A - q_B)x_A x_B + q_A x_B z_A - q_B x_A z_B, \\ \dot{x}_B = (q_B - q_A)x_A x_B - q_A x_B z_A + q_B x_A z_B, \\ \dot{z}_A = 0, \\ \dot{z}_B = 0. \end{cases} \quad (1)$$

Setting $x_B = S - x_A$, $\Delta = q_A - q_B$, reduces (1) to the scalar equation

$$\dot{x}_A = f(x_A) = -\Delta x_A^2 + (\Delta S - q_A z_A - q_B z_B)x_A + q_A z_A S. \quad (2)$$

B.1.1 Equilibria

For $q_A \neq q_B$, the roots of $f(x_A^*) = 0$ are

$$x_{A\pm}^* = \frac{\Delta S - q_A z_A - q_B z_B \pm \sqrt{(q_A z_A + q_B z_B - \Delta S)^2 + 4\Delta q_A z_A S}}{2\Delta}. \quad (3)$$

The physically admissible equilibrium is the root satisfying $0 \leq x_A^* \leq S$, $x_B^* = S - x_A^*$. If $z_A, z_B > 0$, then only x_{A+}^* is feasible. In the symmetric-quality case $q_A = q_B = q$, equation (2) becomes

$$\dot{x}_A = q[z_A S - (z_A + z_B)x_A],$$

and the equilibrium is

$$x_A^* = \frac{z_A}{z_A + z_B} S, \quad x_B^* = \frac{z_B}{z_A + z_B} S. \quad (4)$$

In particular, for equal zealot densities $z_A = z_B = z$,

$$x_A^* = x_B^* = \frac{S}{2}.$$

B.1.2 Stability

Since the reduced model is one-dimensional, local stability is determined by

$$f'(x_A) = -2\Delta x_A + \Delta S - q_A z_A - q_B z_B.$$

For positive zealot densities, the unique internal equilibrium is locally asymptotically stable. Indeed, f is positive at the left boundary and negative at the right boundary, and its unique admissible zero is therefore crossed from positive to negative:

$$f'(x_A^*) < 0.$$

In the symmetric case, this follows directly from

$$f'(x_A^*) = -q(z_A + z_B) < 0.$$

Thus, positive zealot densities remove the neutral continuum of equilibria of the classical symmetric voter model and produce a unique stable coexistence equilibrium. No internal bifurcation occurs as the positive parameters q_A, q_B, z_A, z_B vary.

B.1.3 Boundary case without zealots

When $z_A = z_B = 0$, the model reduces to

$$\dot{x}_A = (q_A - q_B)x_A(1 - x_A).$$

The equilibria are

$$(x_A^*, x_B^*) = (0, 1), \quad (x_A^*, x_B^*) = (1, 0).$$

If $q_A > q_B$, the A -consensus equilibrium $(1, 0)$ is locally asymptotically stable and $(0, 1)$ is unstable. If $q_B > q_A$, their stability is reversed.

When $q_A = q_B$, the deterministic drift vanishes: $\dot{x}_A = 0$. Every point of the interval $x_A \in [0, 1]$ is therefore an equilibrium. This neutral deterministic behaviour differs from the finite stochastic system, in which random fluctuations eventually lead to absorption at one of the two consensus states.

B.2 Voter model with contrarians

The deterministic limit is

$$\begin{cases} \dot{x}_A = (q_A - q_B)x_A x_B + q_A x_B c_A - q_B x_A c_B, \\ \dot{x}_B = (q_B - q_A)x_A x_B - q_A x_B c_A + q_B x_A c_B, \\ \dot{c}_A = -q_A x_A c_A + q_B x_B c_B - q_A c_A^2 + q_B c_B^2, \\ \dot{c}_B = q_A x_A c_A - q_B x_B c_B + q_A c_A^2 - q_B c_B^2. \end{cases} \quad (5)$$

We used that the finite-population correction $X_{C_i}(X_{C_i} - 1)$ has the same limiting density as $X_{C_i}^2$ after normalisation. The system satisfies $\dot{x}_A + \dot{x}_B = 0$, $\dot{c}_A + \dot{c}_B = 0$. Setting $x_B = S - x_A$, $c_B = C - c_A$, reduces (5) to

$$\begin{cases} \dot{x}_A = (q_A - q_B)x_A(S - x_A) + q_A(S - x_A)c_A - q_B x_A(C - c_A), \\ \dot{c}_A = -q_A x_A c_A + q_B(S - x_A)(C - c_A) - q_A c_A^2 + q_B(C - c_A)^2. \end{cases} \quad (6)$$

B.2.1 Equilibria

For general $q_A \neq q_B$, the equilibria are the admissible solutions of the coupled quadratic system

$$\begin{cases} (q_A - q_B)x_A(S - x_A) + q_A(S - x_A)c_A - q_B x_A(C - c_A) = 0, \\ -q_A x_A c_A + q_B(S - x_A)(C - c_A) - q_A c_A^2 + q_B(C - c_A)^2 = 0, \end{cases} \quad (7)$$

subject to $0 \leq x_A \leq S$, $0 \leq c_A \leq C$. These equations can be solved numerically for the asymmetric parameter values used in the experiments. Here we focus on the symmetric benchmark $q_A = q_B = q$, for which the analytical structure is simpler.

Under this assumption, the reduced system becomes

$$\begin{cases} \dot{x}_A = q(S c_A - C x_A), \\ \dot{c}_A = q[C(S + C) - C x_A - (S + 2C)c_A]. \end{cases} \quad (8)$$

At equilibrium, $S c_A^* = C x_A^*$. Substituting this relation and solving, we obtain the unique equilibrium for $C > 0$:

$$x_A^* = x_B^* = \frac{S}{2}, \quad c_A^* = c_B^* = \frac{C}{2}. \quad (9)$$

Thus, in the symmetric voter model, contrarians drive both ordinary agents and contrarians towards an equal split between the two opinions.

B.2.2 Stability

The Jacobian of the symmetric reduced system (8) is constant:

$$J = q \begin{pmatrix} -C & S \\ -C & -(S+2C) \end{pmatrix} \quad (10)$$

Its trace and determinant are

$$\begin{aligned} \text{tr}(J) &= -q(S+3C) < 0, \\ \det(J) &= 2q^2C(S+C) > 0 \end{aligned}$$

for every $C > 0$. Therefore, both eigenvalues have negative real part and the equilibrium (9) is locally asymptotically stable. Since the symmetric reduced system is affine and its Jacobian is constant, the equilibrium is in fact globally asymptotically stable within the invariant physical region $[0, S] \times [0, C]$.

Unlike the cross-inhibition models, the symmetric voter model with positive contrarian density does not exhibit an internal bifurcation: the stable equilibrium remains the equal-split state for every $C > 0$.

B.2.3 Boundary case without contrarians

When $C = 0$, the contrarian compartments disappear and the system reduces to the classical voter model:

$$\dot{x}_A = (q_A - q_B)x_A(1 - x_A).$$

The stability properties are therefore the same as in the no-zealot boundary case (Section B.1.3):

- if $q_A > q_B$, A -consensus is stable;
- if $q_B > q_A$, B -consensus is stable;
- if $q_A = q_B$, every population split is a neutrally stable equilibrium of the deterministic limit.

B.3 Cross-inhibition model with zealots

The deterministic limit is

$$\begin{cases} \dot{x}_A = -q_B x_A x_B + q_A x_A u - q_B x_A z_B + q_A z_A u, \\ \dot{x}_B = -q_A x_A x_B + q_B x_B u - q_A x_B z_A + q_B z_B u, \\ \dot{u} = (q_A + q_B)x_A x_B - (q_A x_A + q_B x_B)u + q_B x_A z_B - q_B u z_B + q_A x_B z_A - q_A u z_A, \\ \dot{z}_A = 0, \\ \dot{z}_B = 0. \end{cases} \quad (11)$$

The system satisfies $\dot{x}_A + \dot{x}_B + \dot{u} = 0$. Hence, setting $u = S - x_A - x_B$ reduces (11) to

$$\begin{cases} \dot{x}_A = q_A(S - x_A - x_B)(x_A + z_A) - q_B x_A(x_B + z_B), \\ \dot{x}_B = q_B(S - x_A - x_B)(x_B + z_B) - q_A x_B(x_A + z_A). \end{cases} \quad (12)$$

B.3.1 Equilibria

At an internal equilibrium, the first two equations give

$$\begin{aligned} q_A u(x_A + z_A) &= q_B x_A(x_B + z_B), \\ q_B u(x_B + z_B) &= q_A x_B(x_A + z_A). \end{aligned}$$

Multiplying these equations and cancelling the positive factors gives $u^2 = x_A x_B$. We introduce

$$t = \sqrt{\frac{x_A}{x_B}} > 0.$$

It follows that $x_A = t^2 x_B$, $u = t x_B$, and more in general

$$x_A = \frac{St^2}{t^2 + t + 1}, \quad x_B = \frac{S}{t^2 + t + 1}, \quad u = \frac{St}{t^2 + t + 1}. \quad (13)$$

Substituting these expressions into the first equilibrium equation gives the cubic equation

$$\Phi(t) = -q_B z_B t^3 + [q_A(S + z_A) - q_B z_B]t^2 + [q_A z_A - q_B(S + z_B)]t + q_A z_A = 0. \quad (14)$$

Every positive root of (14) determines an internal equilibrium through (13). For general asymmetric parameters, the positive roots can be computed numerically.

We now consider the symmetric benchmark $q_A = q_B = q$, $z_A = z_B = z$, $S = 1 - 2z$. In this case, equation (14) factorizes as

$$-q(t-1) [zt^2 + (3z-1)t + z] = 0.$$

The root $t = 1$ always exists and gives the symmetric equilibrium

$$x_A^* = x_B^* = u^* = \frac{S}{3} = \frac{1-2z}{3}. \quad (15)$$

The remaining roots are

$$t_{\pm} = \frac{1-3z \pm \sqrt{(1-z)(1-5z)}}{2z}. \quad (16)$$

They are real, distinct, and positive when

$$0 < z < \frac{1}{5}.$$

For each root $t_{\pm}$, the corresponding asymmetric equilibrium is obtained by substituting $t = t_{\pm}$ into (13). Therefore:

- if $z > \frac{1}{5}$, the symmetric equilibrium is the only internal equilibrium;
- if $z = \frac{1}{5}$, the three equilibrium branches coincide;
- if $0 < z < \frac{1}{5}$, two additional asymmetric internal equilibria exist.

Since z is the density of zealots committed to each option, the critical total zealot density is $2z = 2/5$.

B.3.2 Stability

The Jacobian of the reduced system (12) is

$$J = \begin{pmatrix} q_A(S - 2x_A - x_B - z_A) - q_B(x_B + z_B) & -q_A(x_A + z_A) - q_Bx_A \\ -q_B(x_B + z_B) - q_Ax_B & q_B(S - x_A - 2x_B - z_B) - q_A(x_A + z_A) \end{pmatrix}. \quad (17)$$

In the symmetric case, evaluating the Jacobian at (15) gives

$$J^* = \frac{q}{3} \begin{pmatrix} -(4z+1) & z-2 \\ z-2 & -(4z+1) \end{pmatrix}. \quad (18)$$

Its eigenvalues are

$$\lambda_1 = -q(1+z) < 0, \quad \lambda_2 = \frac{q}{3}(1-5z).$$

Consequently, the symmetric equilibrium is

- locally asymptotically stable if $z > \frac{1}{5}$,
- a saddle point if $0 < z < \frac{1}{5}$.

For an asymmetric equilibrium, equation (16) is equivalently expressed by

$$z = \frac{t}{t^2 + 3t + 1}.$$

Evaluating the trace and determinant of the Jacobian at this equilibrium gives

$$\begin{aligned} \text{tr}(J) &= -\frac{2q(t^2 + t + 1)}{t^2 + 3t + 1} < 0, \\ \det(J) &= \frac{q^2(t-1)^2(t+1)^2}{(t^2 + 3t + 1)^2} > 0 \end{aligned}$$

for $t \neq 1$. Therefore, both asymmetric equilibria are locally asymptotically stable whenever $0 < z < 1/5$.

At $z = 1/5$, the symmetric equilibrium changes stability while the two stable asymmetric equilibria merge with it. The system therefore undergoes a supercritical pitchfork bifurcation.

B.3.3 Boundary case without zealots

When $z_A = z_B = 0$, the model reduces to the ordinary cross-inhibition model:

$$\begin{cases} \dot{x}_A = x_A(q_A u - q_B x_B), \\ \dot{x}_B = x_B(q_B u - q_A x_A), \\ u = 1 - x_A - x_B. \end{cases} \quad (19)$$

The system admits the three boundary equilibria

$$(x_A^*, x_B^*, u^*) \in \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}.$$

It also admits the internal equilibrium

$$x_A^* = \frac{q_B^2}{q_A^2 + q_A q_B + q_B^2}, \quad x_B^* = \frac{q_A^2}{q_A^2 + q_A q_B + q_B^2}, \quad u^* = \frac{q_A q_B}{q_A^2 + q_A q_B + q_B^2}. \quad (20)$$

The two consensus equilibria $(1, 0, 0)$ and $(0, 1, 0)$ are locally asymptotically stable, while the fully undecided equilibrium $(0, 0, 1)$ is unstable. The internal equilibrium (20) is a saddle point.

In the symmetric case $q_A = q_B$, the internal equilibrium becomes

$$x_A^* = x_B^* = u^* = \frac{1}{3}.$$

B.4 Cross-inhibition model with contrarians

The deterministic limit is

$$\begin{cases} \dot{x}_A = -q_B x_A x_B + q_A x_A u - q_B x_A c_B + q_A u c_A, \\ \dot{x}_B = -q_A x_A x_B + q_B x_B u - q_A x_B c_A + q_B u c_B, \\ \dot{u} = (q_A + q_B) x_A x_B - (q_A x_A + q_B x_B) u + q_B x_A c_B - q_B u c_B + q_A x_B c_A - q_A u c_A, \\ \dot{c}_A = -q_A x_A c_A + q_B x_B c_B - q_A c_A^2 + q_B c_B^2, \\ \dot{c}_B = q_A x_A c_A - q_B x_B c_B + q_A c_A^2 - q_B c_B^2. \end{cases} \quad (21)$$

As in the voter model with contrarians, we used that the finite-size correction $X_{C_i}(X_{C_i} - 1)$ has the same limiting density as $X_{C_i}^2$ after normalisation. The system satisfies $\dot{x}_A + \dot{x}_B + \dot{u} = 0$, $\dot{c}_A + \dot{c}_B = 0$, hence setting $u = S - x_A - x_B$, $c_B = C - c_A$, reduces (21) to

$$\begin{cases} \dot{x}_A = q_A u(x_A + c_A) - q_B x_A(x_B + c_B), \\ \dot{x}_B = q_B u(x_B + c_B) - q_A x_B(x_A + c_A), \\ \dot{c}_A = -q_A x_A c_A + q_B x_B c_B - q_A c_A^2 + q_B c_B^2, \end{cases} \quad (22)$$

B.4.1 Equilibria

For general $q_A \neq q_B$, the equilibria are the admissible solutions of the three coupled polynomial equations obtained by setting the right-hand side of (22) equal to zero. These equations can be solved numerically for the asymmetric parameter values used in the experiments. Here we focus on the symmetric case $q_A = q_B = q$.

Like in the previous section, at an internal equilibrium we obtain $u^2 = x_A x_B$ and

$$x_A = \frac{(1 - C)t^2}{t^2 + t + 1}, \quad x_B = \frac{1 - C}{t^2 + t + 1}, \quad u = \frac{(1 - C)t}{t^2 + t + 1}. \quad (23)$$

The first equilibrium equation also gives

$$c_A = \frac{t [Ct^2 + (2C - 1)t + 1]}{(1 + t)(t^2 + t + 1)}, \quad c_B = \frac{t^2 + (2C - 1)t + C}{(1 + t)(t^2 + t + 1)}. \quad (24)$$

Substituting these expressions into $\dot{c}_A = 0$ and simplifying gives

$$(t - 1)(t^2 + Ct + 1) [C(t + 1)^2 - t] = 0. \quad (25)$$

Since $t > 0$ and $C \geq 0$, it holds $t^2 + Ct + 1 > 0$, so this factor produces no admissible roots. The root $t = 1$ gives the symmetric equilibrium

$$x_A^* = x_B^* = u^* = \frac{1 - C}{3} = \frac{S}{3}, \quad c_A^* = c_B^* = \frac{C}{2}. \quad (26)$$

The remaining roots satisfy $C(t+1)^2 - t = Ct^2 + (2C-1)t + C = 0$. For $C > 0$, the roots are

$$t_{\pm} = \frac{1 - 2C \pm \sqrt{1 - 4C}}{2C}. \quad (27)$$

They are real, distinct, and positive if

$$0 < C < \frac{1}{4}.$$

Moreover, $t_+ t_- = 1$, so they describe two symmetric equilibrium branches, one dominated by opinion A and one dominated by opinion B .

At $C = \frac{1}{4}$, the two roots coincide at $t = 1$. Therefore, the critical contrarian density is

$$C_{\text{crit}} = \frac{1}{4}.$$

For an asymmetric equilibrium, the relation $C = \frac{t}{(1+t)^2}$ allows the equilibrium coordinates in (23) and (24) to be written in the simpler form, so for each root $t_{\pm}$, the corresponding asymmetric equilibrium is obtained by substituting $t = t_{\pm}$ into

$$x_A^* = \frac{t^2}{(1+t)^2}, \quad x_B^* = \frac{1}{(1+t)^2}, \quad u^* = \frac{t}{(1+t)^2}, \quad c_A^* = \frac{t}{(1+t)^3}, \quad c_B^* = \frac{t^2}{(1+t)^3}. \quad (28)$$

Therefore:

- if $C > \frac{1}{4}$, the symmetric equilibrium is the only internal equilibrium;
- if $C = \frac{1}{4}$, the asymmetric equilibria coincide with the symmetric one;
- if $0 < C < \frac{1}{4}$, two additional asymmetric internal equilibria exist.

B.4.2 Stability

The Jacobian of the reduced three-dimensional system, evaluated at the symmetric equilibrium (26), is

$$J^* = q \begin{pmatrix} -\frac{3C+S}{3} & -\frac{3C+4S}{6} & \frac{2S}{3} \\ -\frac{3C+4S}{6} & -\frac{3C+S}{3} & -\frac{2S}{3} \\ -\frac{C}{2} & \frac{C}{2} & -\frac{2(3C+S)}{3} \end{pmatrix}. \quad (29)$$

Writing $M = J^*/q$, its characteristic polynomial factorizes as

$$\det(\lambda I - M) = \frac{1}{36} (2\lambda + 3C + 2S) \cdot [18\lambda^2 + (45C + 6S)\lambda + 18C^2 + 6CS - 4S^2]. \quad (30)$$

One eigenvalue is

$$\lambda_1 = -\frac{q}{2}(3C + 2S) < 0.$$

The remaining two eigenvalues have sum and product

$$\begin{aligned} \lambda_2 + \lambda_3 &= -\frac{q}{6}(15C + 2S) < 0 \\ \lambda_2 \lambda_3 &= \frac{q^2}{9}(3C - S)(3C + 2S). \end{aligned}$$

Since $3C + 2S > 0$, the symmetric equilibrium is locally asymptotically stable if and only if

$$3C - S > 0 \iff C > \frac{1}{4}.$$

If instead $0 < C < \frac{1}{4}$, then $\lambda_2 \lambda_3 < 0$, so the symmetric equilibrium is a saddle point.

The two asymmetric equilibria are locally asymptotically stable. Indeed, at an asymmetric equilibrium parametrised by $t \neq 1$, the characteristic polynomial of J/q can be written as

$$\lambda^3 + a_1(t)\lambda^2 + a_2(t)\lambda + a_3(t),$$

where

$$\begin{aligned} a_1(t) &= \frac{3t^6 + 14t^5 + 29t^4 + 36t^3 + 29t^2 + 14t + 3}{(1+t)^6}, \\ a_2(t) &= \frac{3t^6 + 8t^5 + 13t^4 + 15t^3 + 13t^2 + 8t + 3}{(1+t)^6}, \\ a_3(t) &= \frac{(t^3 - 1)^2}{(1+t)^6}. \end{aligned}$$

For $t > 0$ and $t \neq 1$, all three coefficients are strictly positive. Moreover,

$$a_1(t)a_2(t) - a_3(t) = \frac{(t^2 + t + 1)(8t^6 + 20t^5 + 35t^4 + 42t^3 + 35t^2 + 20t + 8)}{(1+t)^8} > 0.$$

The Routh–Hurwitz conditions are therefore satisfied, proving that the two asymmetric equilibria are locally asymptotically stable for $0 < C < 1/4$.

At $C = 1/4$, the symmetric equilibrium changes stability exactly when the two stable asymmetric equilibria merge with it. The system therefore undergoes a supercritical pitchfork bifurcation.

B.4.3 Boundary case without contrarians

When $C = 0$, the contrarian compartments disappear and the system reduces to the ordinary cross-inhibition model analysed in Section B.3.3. The system admits two stable consensus equilibria, one unstable fully undecided equilibrium, and one internal saddle point.